

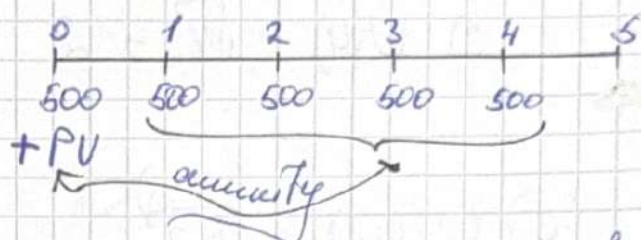
Mini-case 1

• (a) $R = 3,5\% \Rightarrow 0,035$

I. 1) $PV_a = \frac{C}{r} \left[1 - \frac{1}{(1+r)^T} \right]$

$$= \frac{500}{0,035} \left[1 - \frac{1}{(1+0,035)^4} \right] =$$

$$= 1836,54 \text{ — present value of the annuity}$$



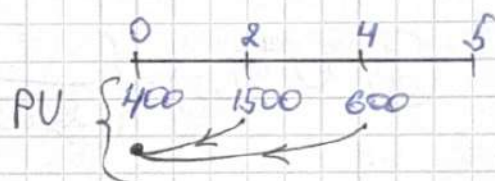
2) $\Sigma PV_1 = PV_a + C_0 = 1836,54 + 500 = 2336,54$

— the sum of the premium payments expressed as DCF (discounted cash flow)

II. 3) $PV_2 = 2500$

III. 4) $PV_3 = \frac{400}{1,035^0} + \frac{1500}{1,035^2} +$

$$+ \frac{600}{1,035^4} = 2323,1318$$



• The most favourable option is the 3rd

• (b) The discounting effect on the future CF would be weakened. Because of it, the PV of the 1st and 3rd options would become greater.

In the 2nd option no interest rate discounts have been employed. Therefore, the PV_2 remains same.

• (c) $P_1 = P_0 (1+r)^t = 2350 (1+0,035)^5 = 2791,0628€$

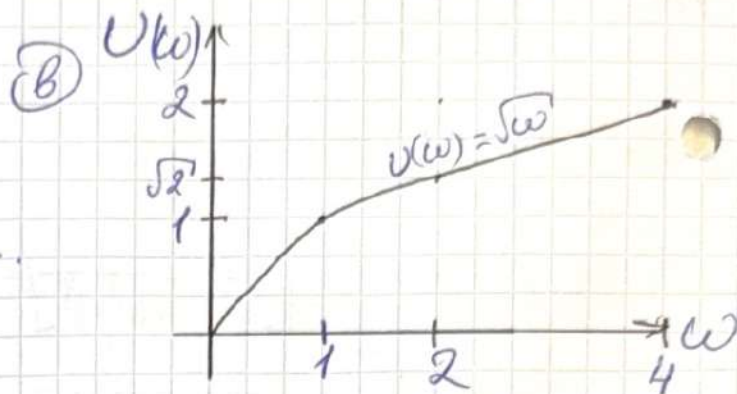
Mini-case 2

a) $U(w) = \sqrt{w}$

1) $U(1) = \sqrt{1} = 1$

2) $U(2) = \sqrt{2} = 1,414...$

3) $U(4) = \sqrt{4} = 2$



c) $U(w) = \sqrt{w}$

$$U'(w) = \left(w^{\frac{1}{2}}\right)' = \frac{1}{2} w^{-\frac{1}{2}} = \frac{1}{2\sqrt{w}}$$

$$\downarrow$$
$$U'(1) = \frac{1}{2\sqrt{1}} = \frac{1}{2} = 0,5$$

$$U'(2) = \frac{1}{2\sqrt{2}} = 0,3535...$$

$$U'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{2 \cdot 2} = \frac{1}{4} = 0,25$$

$$\begin{aligned} d) U''(w) &= \left(\frac{1}{2\sqrt{w}}\right)' = \left[\left(2w^{\frac{1}{2}}\right)^{-1}\right]' = -1 \cdot \left(2w^{\frac{1}{2}}\right)^{-2} \cdot 2 \cdot \frac{1}{2} w^{-\frac{1}{2}} \\ &= -\frac{1}{w^{\frac{1}{2}}(2w^{\frac{1}{2}})^2} = -\frac{1}{w^{\frac{1}{2}} 4w} = -\frac{1}{4w^{\frac{3}{2}}} \end{aligned}$$

The marginal utility is decreasing because of concavity of the utility function ($U''(w) < 0$).

This means the investor is risk averse.

e) 1) $U(w) = 24w$ — risk neutral

$$U'(w) = 24 > 0$$

$$U''(w) = 0$$

2) $U(w) = e^x - 1$ — risk-seeking

$$U'(w) = e^x > 0$$

$$U''(w) = e^x > 0$$
 — convex curve (graph of function)

Mini-case 3

• a) $E_x(W) = 50 \cdot 0,5 + 75 \cdot 0,3 + 150 \cdot 0,2 = 77,5$

$E_y(W) = 25 \cdot 0,6 + 150 \cdot 0,4 = 75$

b) i) X: $U(W) = W^2$

$U(W_1) = 50^2 = 2500$

$U(W_2) = 75^2 = 5625$

$U(W_3) = 150^2 = 22500$

$E\{U(W)\} = 0,5 \cdot 2500 + 0,3 \cdot 5625 + 0,2 \cdot 22500 = 7437,5$

ii) Y: $U(W) = W^2$

$U(W_1) = 25^2 = 625$

$U(W_2) = 150^2 = 22500$

$E\{U(W)\} = 0,6 \cdot 625 + 0,4 \cdot 22500 = 9375$

c) i) If I maximize my wealth, I'll participate in the first lottery (X), since the expected wealth value is greater than the value of Y ($77,5 > 75$)

ii) If I maximize my utility, I prefer to take part in the lottery Y, as the expected utility value in Y is greater than that of the lottery X ($7437,5 < 9375$)

Explanation of the result:

The difference in results between $E(W)$ and $E\{U(W)\}$ can be explained by the player's risk perception.

$U(W) = W^2$; $U'(W) = 2W > 0$; $U''(W) = 2 > 0$;

The utility function tells us that the player (me) is risk seeking. On practice, it means that my marginal utility increases with increasing wealth. And so, greater values of wealth matter for me more than lower ones in the utility valuation.